

## MINOT AFB, ND, USA

WMO: 727675

Lat: 48.417N

Lon: 101.350W

Elev: 508

StdP: 95.37

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94011

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -30.1      | -27.3 | -32.6                         | 0.2 | -29.9 | -29.9 | 0.3 | -26.9 | 15.7                  | -9.9 | 14.5 | -7.9 | 4.6                   | 300  | 0.715 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 7             | 13.2                      | 32.4            | 20.6 | 30.4 | 20.1 | 28.7 | 19.2 | 23.0                | 28.9 | 21.7 | 27.6 | 20.6 | 26.5 | 5.6                     | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.2                            | 16.9 | 26.1 | 19.9 | 15.5 | 24.8 | 18.7 | 14.4 | 23.7 | 70.7          | 29.1 | 65.8 | 27.6 | 61.6 | 26.6 | 29.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 13.5              | 11.9 | 10.6 | -33.6                      | 36.6 | 2.9                | 2.3 | -35.7                                              | 38.2 | -37.4      | 39.6 | -39.0      | 40.8 | -41.2      | 42.5 |    |    |
|                   |      |      | -33.4                      | 25.5 | 2.7                | 1.5 | -35.3                                              | 26.6 | -36.9      | 27.5 | -38.4      | 28.4 | -40.3      | 29.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 4.6    | -13.0 | -10.7 | -3.4  | 4.3   | 12.2  | 17.5  | 20.4  | 19.7  | 14.3  | 6.1   | -2.8  | -10.1 |
|                                                                                  | DBStd       | 13.31  | 8.45  | 8.08  | 7.93  | 5.96  | 4.54  | 3.70  | 3.07  | 3.45  | 4.60  | 5.14  | 6.74  | 7.97  |
|                                                                                  | HDD10.0     | 3087   | 714   | 581   | 416   | 184   | 29    | 2     | 0     | 0     | 13    | 142   | 384   | 623   |
|                                                                                  | HDD18.3     | 5198   | 973   | 814   | 673   | 421   | 195   | 56    | 14    | 25    | 134   | 378   | 634   | 881   |
|                                                                                  | CDD10.0     | 1129   | 0     | 0     | 1     | 13    | 98    | 228   | 321   | 302   | 143   | 22    | 1     | 0     |
|                                                                                  | CDD18.3     | 198    | 0     | 0     | 0     | 0     | 5     | 32    | 77    | 69    | 14    | 1     | 0     | 0     |
|                                                                                  | CDH23.3     | 2657   | 0     | 0     | 0     | 11    | 120   | 424   | 889   | 908   | 287   | 17    | 1     | 0     |
|                                                                                  | CDH26.7     | 906    | 0     | 0     | 0     | 2     | 29    | 122   | 293   | 360   | 96    | 4     | 0     | 0     |
| Wind                                                                             | WSAvg       | 5.0    | 5.5   | 5.3   | 5.3   | 5.4   | 5.4   | 4.8   | 4.1   | 4.1   | 4.6   | 5.2   | 5.1   | 5.2   |
| Precipitation                                                                    | PrecAvg     | 448    | 11    | 11    | 17    | 26    | 73    | 96    | 69    | 51    | 35    | 31    | 17    | 14    |
|                                                                                  | PrecMax     | 587    | 36    | 90    | 34    | 59    | 169   | 255   | 199   | 100   | 105   | 147   | 69    | 41    |
|                                                                                  | PrecMin     | 296    | 1     | 0     | 3     | 8     | 16    | 37    | 28    | 14    | 7     | 1     | 0     | 3     |
|                                                                                  | PrecStd     | 87     | 9     | 17    | 10    | 13    | 41    | 41    | 35    | 23    | 24    | 30    | 17    | 9     |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 5.9   | 8.0   | 18.8  | 24.9  | 30.2  | 32.7  | 34.5  | 36.0  | 32.8  | 25.4  | 17.0  | 8.8   |
|                                                                                  |             | MCWB   | 2.9   | 4.1   | 10.1  | 12.9  | 17.4  | 21.1  | 22.6  | 20.7  | 19.6  | 14.4  | 9.6   | 4.3   |
|                                                                                  | 2%          | DB     | 3.6   | 4.2   | 14.0  | 20.5  | 26.4  | 30.0  | 31.7  | 32.7  | 29.2  | 21.3  | 12.3  | 5.1   |
|                                                                                  |             | MCWB   | 1.6   | 2.3   | 8.0   | 10.7  | 15.3  | 19.5  | 22.0  | 20.1  | 17.3  | 12.8  | 6.8   | 2.0   |
|                                                                                  | 5%          | DB     | 1.8   | 2.5   | 10.0  | 17.6  | 24.0  | 27.6  | 29.9  | 30.5  | 27.0  | 18.0  | 8.9   | 2.8   |
|                                                                                  |             | MCWB   | 0.4   | 1.0   | 6.0   | 9.4   | 13.9  | 18.3  | 21.4  | 19.4  | 16.5  | 10.7  | 4.8   | 0.6   |
|                                                                                  | 10%         | DB     | -0.5  | 0.7   | 6.3   | 14.8  | 22.0  | 25.8  | 27.9  | 28.3  | 23.9  | 14.8  | 6.1   | 0.7   |
|                                                                                  |             | MCWB   | -1.5  | -0.3  | 3.8   | 8.0   | 13.1  | 17.5  | 20.1  | 18.9  | 15.3  | 9.5   | 3.2   | -0.8  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 3.5   | 4.7   | 11.0  | 13.8  | 19.0  | 23.7  | 25.5  | 23.3  | 21.4  | 16.1  | 10.2  | 4.5   |
|                                                                                  |             | MCDB   | 5.5   | 7.4   | 17.2  | 23.1  | 26.3  | 29.2  | 30.2  | 30.2  | 29.2  | 22.7  | 16.6  | 8.2   |
|                                                                                  | 2%          | WB     | 1.8   | 2.5   | 8.6   | 11.6  | 17.1  | 21.3  | 23.7  | 21.9  | 19.3  | 13.5  | 7.3   | 2.3   |
|                                                                                  |             | MCDB   | 3.4   | 4.2   | 13.2  | 18.7  | 23.5  | 26.5  | 29.3  | 28.5  | 26.2  | 19.5  | 11.5  | 4.8   |
|                                                                                  | 5%          | WB     | 0.4   | 1.0   | 6.0   | 10.0  | 15.4  | 19.8  | 22.4  | 20.9  | 17.7  | 11.7  | 5.1   | 0.8   |
|                                                                                  |             | MCDB   | 1.8   | 2.3   | 9.7   | 16.6  | 21.8  | 25.4  | 27.8  | 27.3  | 23.5  | 16.5  | 8.3   | 2.9   |
|                                                                                  | 10%         | WB     | -1.6  | -0.2  | 3.9   | 8.5   | 14.0  | 18.5  | 21.2  | 19.9  | 16.2  | 10.0  | 3.4   | -0.8  |
|                                                                                  |             | MCDB   | -0.6  | 0.6   | 6.4   | 13.8  | 19.9  | 24.2  | 26.7  | 26.5  | 22.4  | 14.6  | 5.9   | 0.6   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.9   | 9.8   | 10.4  | 12.7  | 13.4  | 12.4  | 13.2  | 14.9  | 14.1  | 11.9  | 10.1  | 9.3   |
|                                                                                  |             | MCDBR  | 11.0  | 10.7  | 15.6  | 18.1  | 17.3  | 15.9  | 15.6  | 18.4  | 19.2  | 17.7  | 15.1  | 11.2  |
|                                                                                  |             | MCWBR  | 9.4   | 8.7   | 10.0  | 9.9   | 8.3   | 7.8   | 7.5   | 7.8   | 8.9   | 9.9   | 10.0  | 8.7   |
|                                                                                  | 5% WB       | MCDBR  | 10.9  | 10.1  | 15.0  | 16.8  | 15.0  | 13.5  | 14.1  | 15.4  | 16.2  | 16.3  | 14.0  | 11.1  |
|                                                                                  |             | MCWBR  | 9.5   | 8.4   | 9.9   | 9.7   | 8.0   | 7.6   | 7.7   | 7.6   | 8.4   | 9.7   | 9.5   | 8.9   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.250  | 0.257 | 0.291 | 0.329 | 0.345 | 0.366 | 0.376 | 0.384 | 0.341 | 0.298 | 0.265 | 0.243 |       |
|                                                                                  | taud        | 2.402  | 2.424 | 2.450 | 2.400 | 2.408 | 2.394 | 2.382 | 2.337 | 2.428 | 2.526 | 2.494 | 2.447 |       |
|                                                                                  | Ebn at Noon | 843    | 920   | 934   | 923   | 916   | 895   | 880   | 854   | 864   | 858   | 821   | 811   |       |
|                                                                                  | Edh at Noon | 64     | 79    | 92    | 108   | 112   | 115   | 114   | 115   | 95    | 72    | 59    | 54    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.58   | 2.69  | 3.90  | 4.89  | 5.76  | 6.20  | 6.60  | 5.40  | 3.99  | 2.50  | 1.50  | 1.11  |       |
|                                                                                  | RadStd      | 0.14   | 0.26  | 0.32  | 0.37  | 0.40  | 0.44  | 0.26  | 0.38  | 0.29  | 0.27  | 0.22  | 0.12  |       |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) | Regional (11 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |

Nomenclature: See separate page